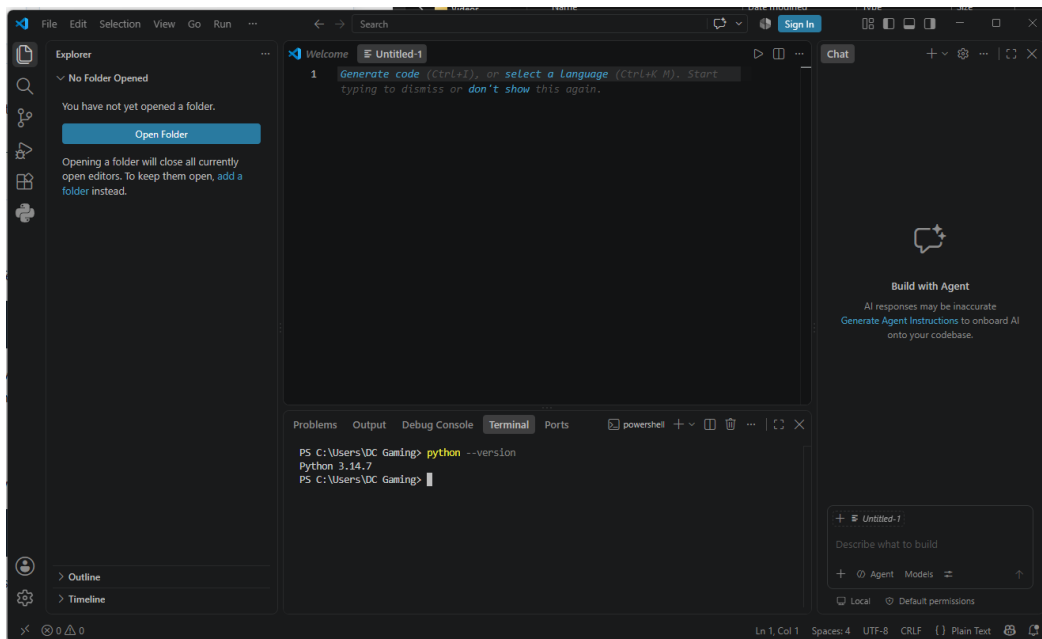
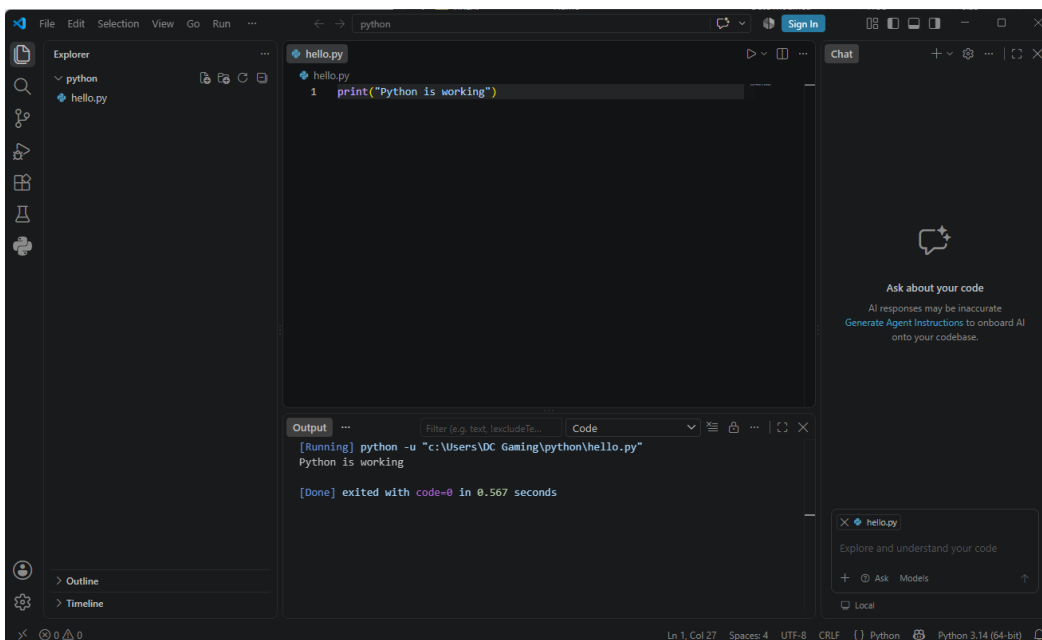


Part E – Evidence

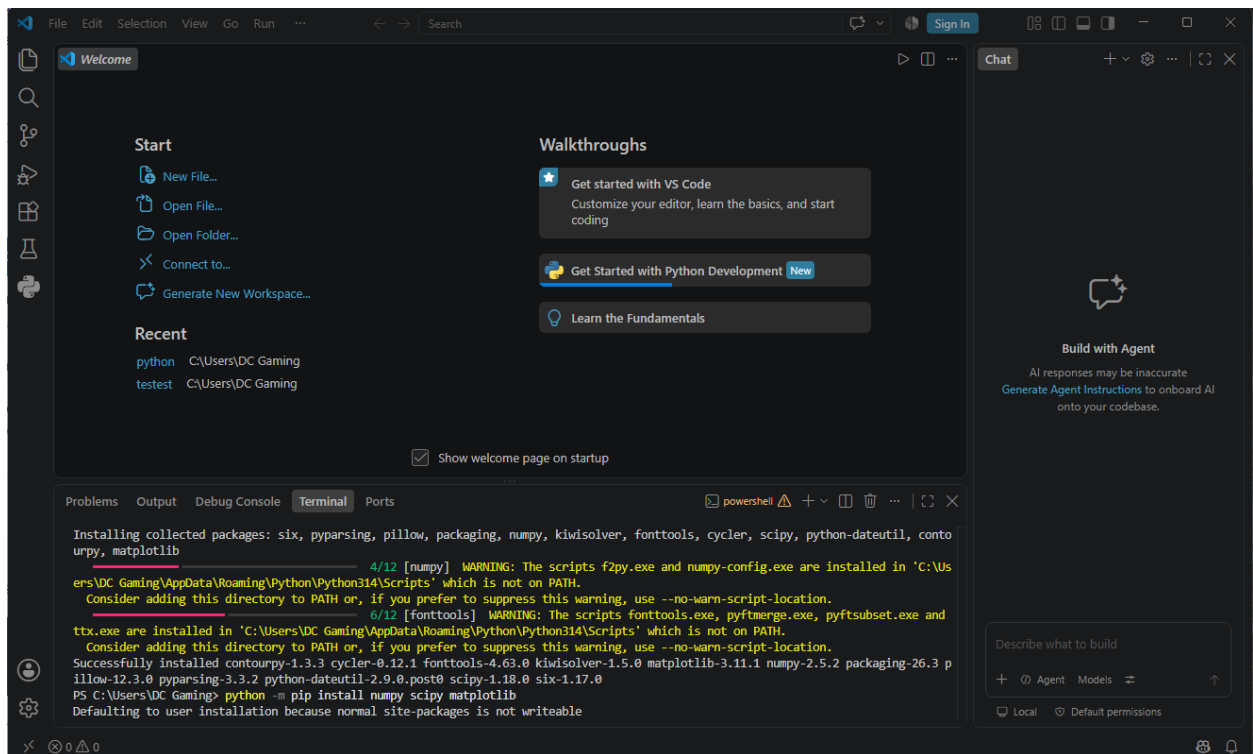
1. The terminal showing `python --version` and its answer.



2. VS Code with `hello.py` open and its output visible in the terminal.



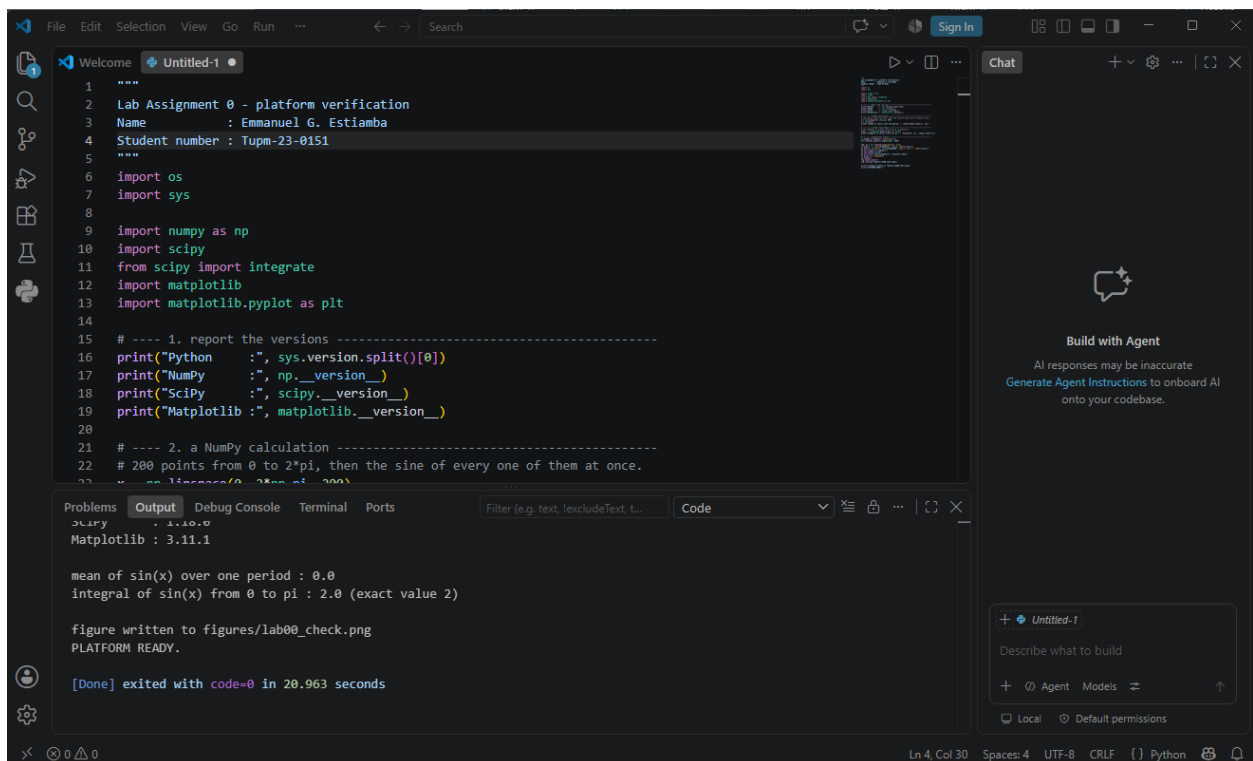
3. The terminal showing the successful installation of NumPy, SciPy and Matplotlib.



The screenshot shows the Visual Studio Code interface with the terminal window open. The terminal displays the output of a pip install command, showing the successful installation of NumPy, SciPy, and Matplotlib. The output includes warnings about the installation location and a confirmation message.

```
Installing collected packages: six, pyparsing, pillow, packaging, numpy, kiwisolver, fonttools, cycler, scipy, python-dateutil, contourpy, matplotlib
  WARNING: The scripts f2py.exe and numpy-config.exe are installed in 'C:\Users\DC Gaming\AppData\Roaming\Python\Python314\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
  WARNING: The scripts fonttools.exe, pyftmerge.exe, pyftsubset.exe and tttx.exe are installed in 'C:\Users\DC Gaming\AppData\Roaming\Python\Python314\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed contourpy-1.3.3 cycler-0.12.1 fonttools-4.63.0 kiwisolver-1.5.0 matplotlib-3.11.1 numpy-2.5.2 packaging-26.3 pyparsing-3.3.2 python-dateutil-2.9.0.post0 scipy-1.18.0 six-1.17.0
PS C:\Users\DC Gaming> python -m pip install numpy scipy matplotlib
Defaulting to user installation because normal site-packages is not writeable
```

4. The full console output of your verification program, ending in **PLATFORM READY.**



The screenshot shows the Visual Studio Code interface with a Python script named 'Untitled-1' open. The script performs a platform verification by importing NumPy, SciPy, and Matplotlib, and then calculating the mean of sin(x) over one period and the integral of sin(x) from 0 to pi. The output of the script is displayed in the terminal window, showing the versions of the installed packages and the results of the calculations.

```
1 """
2 Lab Assignment 0 - platform verification
3 Name      : Emmanuel G. Estiamba
4 Student number : Tupm-23-0151
5 """
6 import os
7 import sys
8
9 import numpy as np
10 import scipy
11 from scipy import integrate
12 import matplotlib
13 import matplotlib.pyplot as plt
14
15 # ---- 1. report the versions ----
16 print("Python      :", sys.version.split()[0])
17 print("NumPy       :", np.__version__)
18 print("SciPy        :", scipy.__version__)
19 print("Matplotlib    :", matplotlib.__version__)
20
21 # ---- 2. a NumPy calculation ----
22 # 200 points from 0 to 2*pi, then the sine of every one of them at once.
23 x = np.linspace(0, 2*np.pi, 200)
```

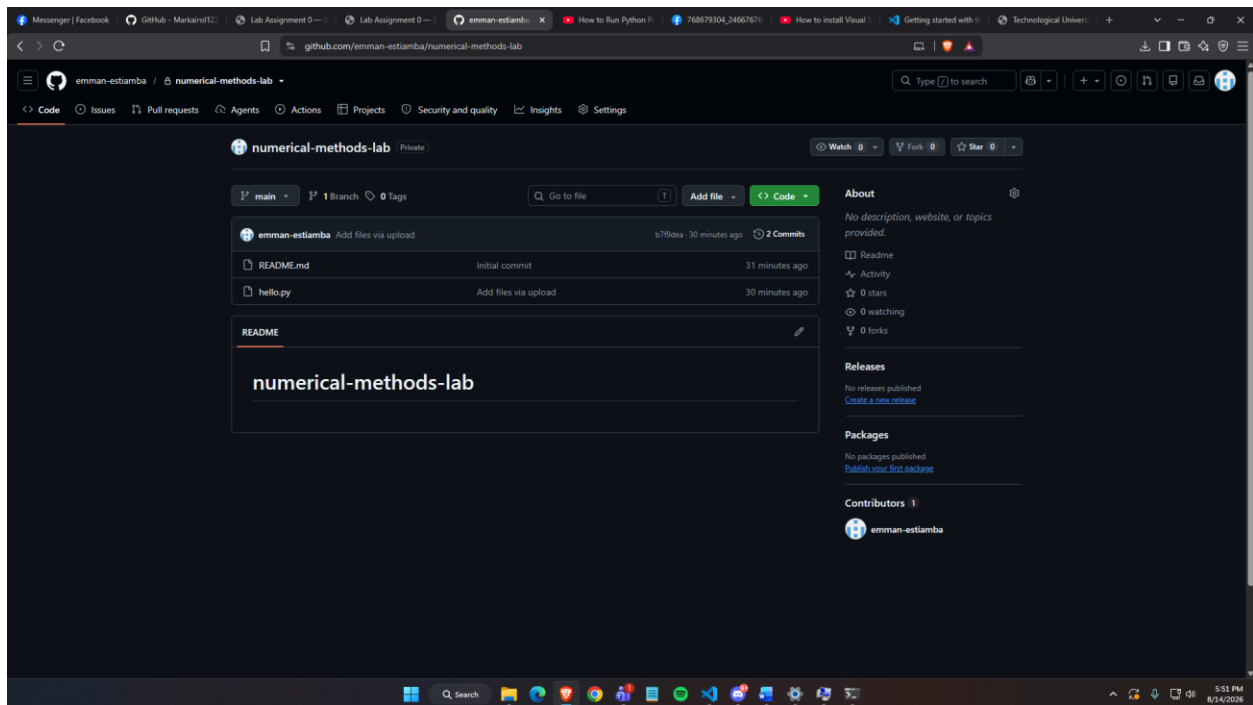
```
Python      : 3.10.0
NumPy       : 1.24.0
SciPy        : 1.10.1
Matplotlib    : 3.11.1

mean of sin(x) over one period : 0.0
integral of sin(x) from 0 to pi : 2.0 (exact value 2)

figure written to figures/lab00_check.png
PLATFORM READY.

[Done] exited with code=0 in 20.963 seconds
```

5. Your GitHub repository page, showing the repository name and the uploaded file.



- Figures/lab00_check.png

